

PEPTIDE DISCOVERY USING BACTERIAL DISPLAY AND FLOW CYTOMETRY

Jennifer A. Getz, Tobias D. Schoep, *and* Patrick S. Daugherty

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Abstract

Peptides are increasingly used as therapeutic and diagnostic agents. The combination of bacterial cell-surface display peptide libraries with magnetic- and fluorescence-activated cell sorting technologies provides an efficient and highly effective methodology to identify and engineer peptides for a growing number of molecular recognition applications. Here, detailed protocols for both the generation and screening of bacterial display peptide libraries are presented. The methods described enable the discovery and evolutionary optimization of protein-binding peptides, cell-specific peptides, and enzyme substrates for diverse biotechnology applications.

Department of Chemical Engineering, Institute for Collaborative Biotechnologies, University of California, Santa Barbara, California, USA

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1. INTRODUCTION

1.1. Principles of peptide isolation using bacterial display libraries

Biological display technologies have been applied to identify peptides that bind to various targets including proteins, whole cells, and synthetic materials. Bacterial surface display is a well-established methodology for the discovery and optimization of peptides with desired functions for an expanding group of applications. Typically, bacteria that display target-binding peptides are enriched from large libraries exceeding one billion members using sequential magnetic-activated cell sorting (MACS) and fluorescence-activated cell sorting (FACS) (Fig. 4.1). Thus, bacterial display peptide libraries are distinct from other peptide library technologies, including bacteriophage display, in that FACS instrumentation allows real-time analysis of the properties of peptides displayed on the cell surface during screening; measurable properties include target-binding affinity and specificity, and peptide stability to proteases. The ability to measure these properties and tune the stringency of the screen enables efficient identification of peptides with desired properties for a wide range of purposes (Daugherty, 2007). In particular, bacterial display libraries have been used to identify peptides with potential utility in therapeutic or diagnostic applications (Table 4.1). Here, we describe detailed methods that enable the selection of peptides from bacterial display libraries that bind or otherwise interact with a target.

1.2. Bacterial peptide display scaffolds

A variety of display scaffolds have been exploited to present peptides on the surface of bacteria. Scaffolds have been developed for both Gram-negative and Gram-positive bacteria, although Gram-negative based systems have been more extensively utilized (Jostock and Dubel, 2005). General considerations involved in bacterial display and library screening have been reviewed (Daugherty, 2007; Strauch and Georgiou, 2009), and improvements to display scaffolds and screening methods are rapidly evolving (Binder *et al.*, 2010; Chen *et al.*, 2008; Kronqvist *et al.*, 2008). Peptides can be displayed on the surface of *Escherichia coli* by: insertion into outer membrane proteins (e.g., OmpA, OmpX) (Besette *et al.*, 2004) or flagellar proteins (FliC) (Lu *et al.*, 1995); fusion to surface exposed N-termini (e.g., IgA protease) (Wentzel *et al.*, 1999); or fusion to both surface exposed N- and C-termini (e.g., circularly permuted OmpX (CPX)) (Rice *et al.*, 2006). The use of terminally fused peptide libraries is especially useful for identifying peptides that are more likely to retain function when removed from

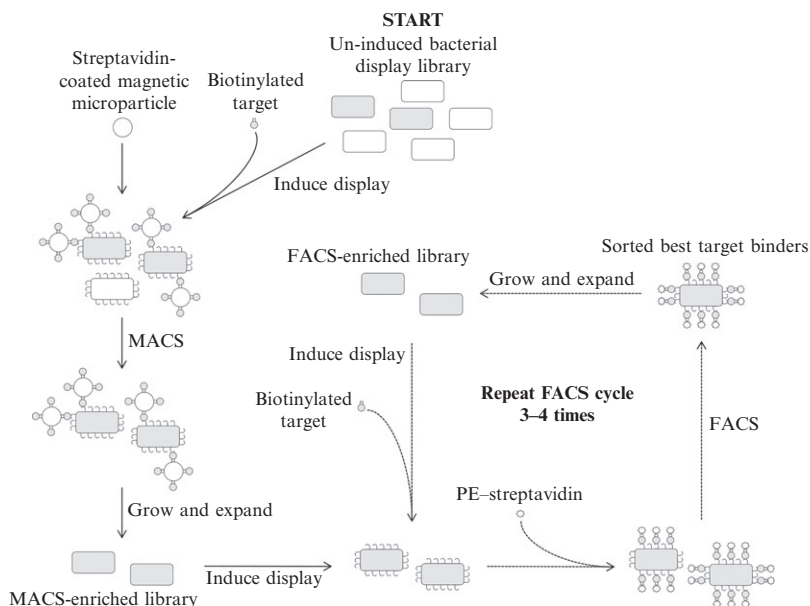


Figure 4.1 Identification of target-binding peptides from bacterial display peptide libraries. Peptide-displaying bacteria are incubated with biotinylated target and streptavidin-functionalized magnetic microparticles, and selected using MACS. The recovered cells are amplified by growth and then screened for binding to labeled targets using FACS. Additional cycles of FACS and regrowth are performed (dotted lines) to enrich bacteria that display peptides with the desired affinity and specificity. PE, phycoerythrin.

their scaffold. Each display scaffold, however, imposes differing constraints upon passenger peptides. Consequently, the methods presented are specific to the host bacterial strain and display scaffold.

Peptides can be displayed in a non-constrained fashion, or in a constrained conformation with one or more stabilizing disulfide bonds. Typically, constrained peptides exhibit increased binding affinity and specificity (Getz *et al.*, 2011; Smith and Petrenko, 1997) since they can adopt fewer conformations when compared to linear peptides. Additionally, the use of structural constraints may better preserve peptide-binding properties when peptides are characterized independent of the display scaffold. Structurally constrained peptide libraries have proven superior for generating high-affinity and high-stability peptides in several studies (Getz *et al.*, 2011; Orning *et al.*, 2006; Uchiyama *et al.*, 2005). Nevertheless, binders exhibiting disulfide constraints can often be directly identified from fully random, non-constrained libraries (Kenrick and Daugherty, 2010; Rojas *et al.*, 2008).

Table 4.1 Applications of bacterial display peptide libraries

Achievement	Potential use	Organism	Scaffold	Example reference
Identification of tumor cell-specific peptides	Tumor targeted delivery	<i>E. coli</i>	FliTrx	Zitzmann <i>et al.</i> (2005) and Li <i>et al.</i> (2008)
Identification of breast tumor cell-specific peptides		<i>E. coli</i>	OmpX (CPX)	Dane <i>et al.</i> (2009)
Identification of high-affinity VEGF-binding peptides	Peptide therapeutic agonists/antagonists	<i>E. coli</i>	OmpX (eCPX)	Kenrick and Daugherty (2010)
Identification of high-affinity TNF α -binding affibodies		<i>Staphylococcus carnosus</i>	Protein A (fragment)	Kronqvist <i>et al.</i> (2008)
Identification of high-affinity CTLA-4-binding anticalin		<i>E. coli</i>	EspP autotransporter/lipocalin	Binder <i>et al.</i> (2010)
Epitope mapping of antigens		<i>S. carnosus</i>	Protein A (fragment)	Rockberg <i>et al.</i> (2008)
Identification of protease-activated proligands	Pathology mediated activation of peptide based therapeutics	<i>E. coli</i>	OmpX (eCPX)	Thomas and Daugherty (2009)

1.3. Bacterial peptide display systems

Various display systems have been used to generate peptides that bind a target molecule (Jostock and Dubel, 2005; van Bloois *et al.*, 2011). A majority of previously reported bacterial display scaffolds and libraries are not available commercially and must be obtained by request or reconstructed. The protocols presented below are specific to peptide libraries displayed using the eCPX scaffold (Rice and Daugherty, 2008). The eCPX scaffold is a circularly permuted variant of the outer membrane protein OmpX (Rice *et al.*, 2006) with additional mutations that aid passenger presentation on the cell surface (Rice and Daugherty, 2008). Peptide libraries can be displayed at either the N- or C-terminus of the protein and can be utilized for diverse applications (Fig. 4.2). We and others have demonstrated the use of eCPX to identify tumor cell-specific peptides (Dane *et al.*, 2009), generate and affinity mature protein-binding peptides (Kenrick and Daugherty, 2010; Sun *et al.*, 2009), construct protease-activated VEGF-binding ligands (Thomas and Daugherty, 2009), and identify determinants of antibody-epitope binding specificity (Hall and Daugherty, 2009).



2. PROTOCOLS

2.1. Construction of large bacterial display peptide libraries

The following protocol provides detailed instructions for constructing bacterial display libraries with the eCPX display scaffold (Rice and Daugherty, 2008). The pBAD33-eCPX plasmid (available from Addgene) includes an arabinose operon (araBAD) promoter and the chloramphenicol acetyltransferase gene to confer antibiotic resistance to chloramphenicol (CM) (Fig. 4.3). The following sections describe the preparation of the vector and insert DNA, performing test ligations, library construction, and production of electrocompetent *E. coli* cells.

2.1.1. Preparation of vector DNA

1. Isolate at least 10 μg of pBAD33-eCPX plasmid at a concentration of approximately 200 ng/ μL .
2. Digest the vector DNA with 10 units of *Sfi*I restriction enzyme per microgram of DNA in a 200- μL reaction volume using the appropriate buffer and BSA concentration. Incubate at 50°C for 4h.
3. Prepare a 0.7% (w/v) agarose gel; rinse and fill the gel box with TAE buffer. Load DNA ladder into lane 1 and the digestion reaction in the remaining lanes (less than 5 μg each) and run for 45min at 100V. SYBER Safe DNA gel stain (Invitrogen) can be used with a blue light

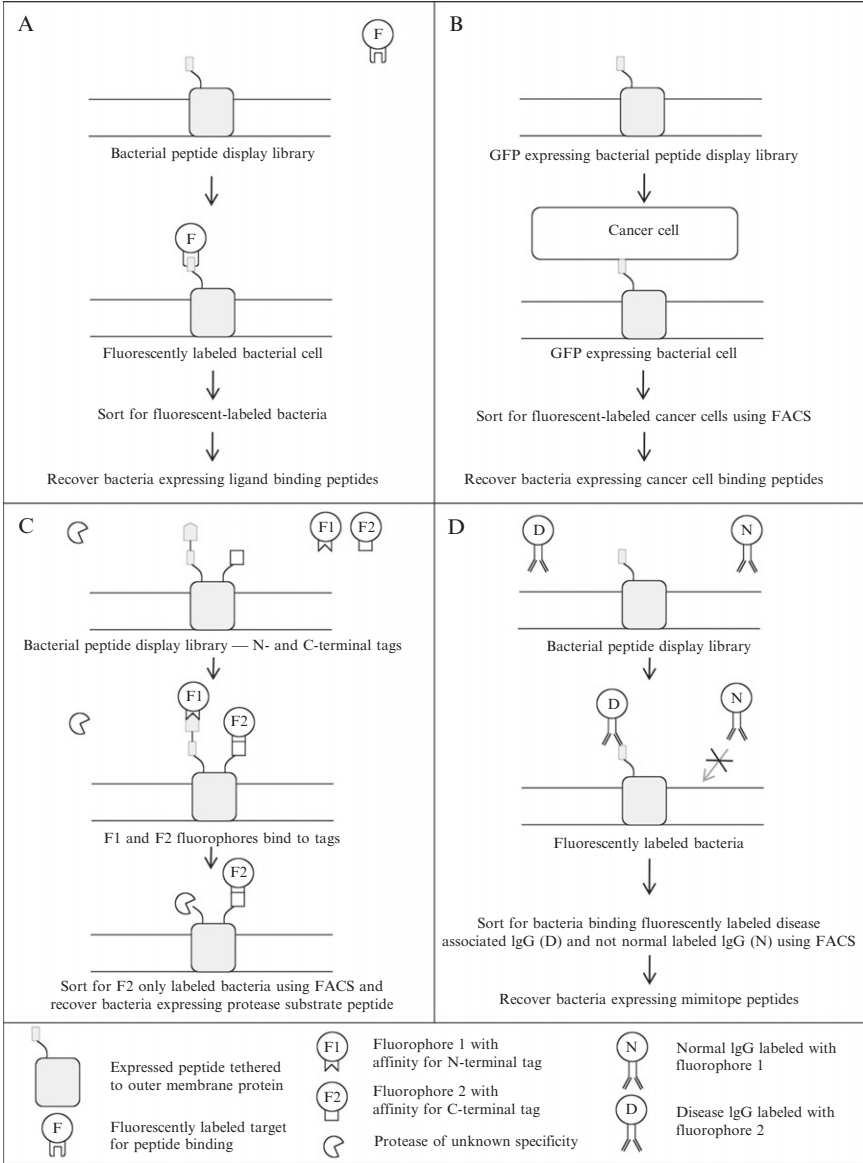


Figure 4.2 Bacterial peptide surface display and analysis strategies. (A) Identification of peptide binders to a target protein. (B) Identification of peptides that bind cultured cells or tissue. (C) Identification of protease substrates using cellular libraries of peptide substrates (CLiPS). (D) Identification of peptides (mimotopes) that bind disease-associated antibodies.

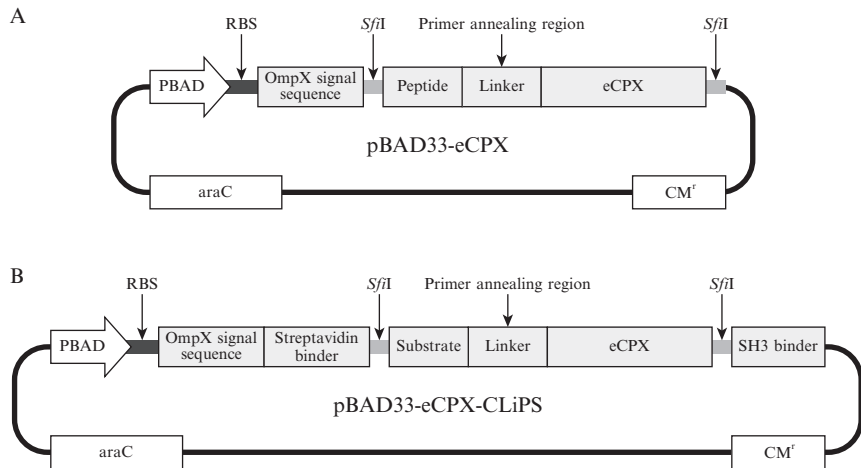


Figure 4.3 Plasmids for bacterial display. (A) Map of plasmid pBAD33-eCPX for expression of peptide libraries fused to the N-terminus of the display scaffold eCPX. The plasmid carries a chloramphenicol resistance marker (CM') and an arabinose promoter (PBAD). (B) Map of plasmid pBAD33-eCPX-CLiPS for expression of substrate libraries with tags on both the N- and C-termini of eCPX.

transilluminator to reduce nicking caused by the traditional use of ethidium bromide and UV light.

4. Excise the bands corresponding to the double-cut plasmid (~5400bp) using a razor blade; extract and purify the DNA using the Zymoclean gel DNA recovery kit (each column can bind up to 5 μ g of DNA); and elute with a total volume (across all columns) of 50 μ L of sterile MilliQ water.
5. For incompletely digested vector DNA, dephosphorylation of the vector can prevent ligation of single-cut plasmid and therefore reduce the amount of template in the library. Add 1/10 volume of 10 $\times$ Antarctic Phosphatase buffer (NEB) and 1 μ L (5 units) of Antarctic Phosphatase per 1–5 μ g DNA. Incubate at 37°C for 60 min and heat inactivate the enzyme for 5 min at 65°C.

2.1.2. Preparation of insert DNA

To construct a random library on the N-terminus of eCPX, oligonucleotide primers can be designed that anneal to DNA coding the short linker region (GGQSGQ) upstream of eCPX and extend to the *Sfi*I site immediately downstream of the OmpX signal sequence (Fig. 4.3). The reverse primer anneals downstream of the *Sfi*I site on the C-terminal end of eCPX. Degenerate codons of the form NNS (N=A,G,C,T; S=G,C) are utilized

to create the randomized region. The forward and reverse primers used to construct a 15-mer peptide library are listed below:

Forward primer. ACTTCCGTAGCTGGCCAGTCTGGCCAGNNS
NNSNNSNNSNNSNNSNNSNNSNNSNNSNNS
NNSNNSNNSNNSGGAGGGCAGTCTGGGCAG
TCTG

Reverse primer. GGCTGAAAATCTTCTCTC

1. Dilute the oligonucleotide primers to 10 μ M.
2. For the PCR reaction, mix the following and then aliquot into 10 0.2 mL PCR tubes:
325 μ L sterile MilliQ water
50 μ L 10 $\times$ DNA polymerase buffer
5 μ L diluted vector DNA (~100 ng total of plasmid)
15 μ L forward (5') primer (10 μ M stock)
15 μ L reverse (3') primer (10 μ M stock)
50 μ L dNTPs (2 mM each)
30 μ L MgSO₄ (25 mM)
10 μ L proofreading, hot start DNA polymerase, 1 unit/ μ L (KOD Hot-Start Polymerase, EMD Chemicals)
3. Perform a PCR with the following conditions:
 - a. 94°C for 2 min
 - b. 94°C for 30 s
 - c. 55°C for 30 s
 - d. 68°C for 1 min
 - e. Repeat steps b through d for 25 cycles
4. Check the insert PCR product (~600–700 bp) by analyzing 5 μ L of the reaction on an agarose gel. Purify the remaining reaction using the Zymo DNA Clean and Concentrator kit. The PCR should yield approximately 10 μ g of insert DNA.
5. Digest 10 μ g (or the amount recovered) of insert DNA with 10 units of *Sfi*I per microgram of DNA for 4 h at 50°C, purify using the Zymo DNA Clean and Concentrator kit (each column can bind up to 5 μ g of DNA), and elute with a total volume (across all columns) of 50 μ L of sterile MilliQ water.

2.1.3. Performing test ligations

A protocol for making high efficiency electrocompetent *E. coli* cells is described at the end of [Section 2.1.4](#). The construction of large libraries is dependent upon electrocompetent cells with a transformation efficiency of at least 1×10^{10} CFU/ μ g pUC18.

1. Determine the concentration of the digested vector and insert DNA. To do this, dilute the DNA 1:50 and measure the absorbance at 260 and 280 nm (the absorbance reading should be in the linear range between 0.01 and 1). Then, calculate the DNA concentration using the following conversion: $A_{260 \text{ nm}}$ of 1.0 = 50 ng/μL DNA. The DNA purity, as determined by the ratio of absorbance at 260 nm versus 280 nm, should be greater than 1.8. With perfect recovery, the vector and insert concentrations should be approximately 200 ng/μL, but concentrations as low as 50 ng/μL are sufficient.
2. For the test ligations, try multiple molar ratios of insert to vector ($I:V$) in the reactions: 0:1 (vector only control), 1:1, 3:1, and 5:1. Use the same mass of vector DNA (~50 ng) for each ligation reaction and adjust the amount of insert to achieve the desired ratio. Also, make sure to add less than 100 ng of total DNA (insert plus vector) per 20 μL ligation reaction. Equation (4.1) can be used to calculate the amount of insert DNA for each reaction:

$$\text{ng}_V \times \frac{\text{bp}_I}{\text{bp}_V} \times I:V = \text{ng}_I \quad (4.1)$$

where ng_V is the mass (ng) of vector in the ligation reaction; bp_I and bp_V are the number of base pairs in the insert and vector DNA, respectively; $I:V$ is the ratio of insert to vector; and ng_I is the mass (ng) of insert.

3. Each test ligation reaction should have a volume of 20 μL with the following components: 4 μL of 5× T4 DNA ligase buffer, vector DNA, insert DNA, sterile MilliQ water, and 1 μL of T4 DNA ligase (Invitrogen).
4. Incubate ligation reactions at 14°C for 16 h and heat inactivate the ligase by incubation at 70°C for 10 min.
5. Desalt 10 μL of the ligation reactions using a membrane filter (Millipore “V” Series, 0.025 μm) on sterile MilliQ water for 2 h. The reaction volume on the surface of the filter will roughly double in this time period.
6. Thaw four 70 μL aliquots of electrocompetent MC1061 cells on ice. Add 10 μL of each chilled, desalted ligation to the cell aliquots and transfer to an ice cold 1 mm electroporation cuvette. Pulse at 1.8 kV, 50 μF, and 100 Ω (Bio-Rad Gene Pulser) and rinse the cuvette twice with 0.75 mL of warm SOC media to recover the bacteria and incubate with shaking at 37°C for 1 h.
7. Plate 100 μL of 1:10², 1:10³, and 1:10⁴ dilutions of each of the transformations (1:10 and 1:10² for the vector only control transformation) on LB-agar plates supplemented with 34 μg/mL CM as the selection antibiotic.
8. Incubate overnight at 37°C and on the following day, count the number of colonies on each of the plates to calculate the total number of transformants. For example, 100 CFU on the 1:10³ dilution plate would equal

1.2×10^8 CFU/ μ g (assuming 12.5 ng vector DNA per transformation). The transformation efficiency should be between 10^7 and 10^9 transformants/ μ g of vector DNA. Determine which insert to vector ratio in the ligation produces the maximum number of transformants. Also, the number of transformants on the vector-only control plate divided by the transformants under optimum ligation conditions can be used to estimate the fraction of background or template DNA that will result in the library. If the library background is greater than 10%, redo the vector digestion with an increased enzyme concentration and use Antarctic Phosphatase to prevent the ligation of single-cut plasmid.

2.1.4. Library construction

1. Using the best insert to vector ratio, scale-up the volume of the ligation reaction (e.g., by 50–100 $\times$) depending on the size of the desired library and the amount of digested insert and vector DNA available. Using 5–10 μ g of vector DNA in the ligation should be adequate.
2. Divide the larger reaction volume into 100 μ L aliquots, incubate for 16 h at 14 $^{\circ}$ C, and deactivate the ligase by heating to 70 $^{\circ}$ C for 10 min.
3. To reduce the volume after ligation, precipitate the DNA (divided into 400 μ L aliquots) with 1/10 the volume of 3 M sodium acetate (pH 5.2), 2 μ L oyster glycogen (20 mg/mL), and 2.2 volumes of ethanol (alternatively use Novagen Pellet Paint Co-Precipitant, or use the Zymo DNA Clean and Concentrator kit to skip the precipitation).
4. Mix and incubate for 1–2 h at –80 $^{\circ}$ C or overnight at –20 $^{\circ}$ C.
5. Centrifuge at maximum speed for 15 min at 4 $^{\circ}$ C. Remove the supernatant and allow the pellet to dry until the edges begin to turn translucent.
6. Resuspend the pellet in 50 μ L of sterile MilliQ water.
7. Desalt the ligated DNA in $\sim$ 10–15 μ L aliquots using membrane filters floating on sterile MilliQ water for 2 h.
8. Chill aliquots of 0.5–3 μ g of desalted DNA on ice (the volume should be <30 μ L). Thaw four or more (depending on the number of transformations) 250 μ L aliquots of electrocompetent cells on ice and add the desalted DNA as soon as the competent cells are thawed. Transfer the mixture to an ice cold 2 mm electroporation cuvette and pulse at 2.5 kV, 50 μ F, and 100 Ω . Rinse the cuvette three times with 1 mL of warm SOC media and transfer the cells to a 125-mL flask containing 7 mL of SOC (10 mL total volume per flask).
9. Repeat procedure for the remaining transformations and incubate the flasks with shaking at 37 $^{\circ}$ C for 1 h.
10. If all the time constants during electroporation were approximately the same, pool the transformations into a single flask. Plate 100 μ L of 1:10³, 1:10⁴, and 1:10⁵ dilutions on LB-agar/CM plates. Repeat the series

of dilutions and prepare duplicate plates. Incubate the plates overnight at 37°C.

11. Add the pooled transformations to a 2-L flask containing 500mL LB/CM with 0.2% (w/v) sterile-filtered glucose. Incubate the flask at 37°C until the culture reaches an OD₆₀₀ of 2 (3–8h depending on library size).
12. Inoculate a sterile 2-L flask containing 400mL LB/CM and 0.2% glucose with 100mL of the transformation culture and grow overnight.
13. Centrifuge the remaining 400mL of culture at 3000×g for 15 min.
14. Resuspend the pellet in 10mL SOB supplemented with 15% (v/v) glycerol. Aliquot into cryovials and freeze at –80°C. Each frozen stock should contain at least 10 times the number of independent transformants.
15. The following morning, count the colonies on the plates to determine the library size (number of independent transformants).
16. Isolate plasmid DNA from the overnight culture of the newly constructed library using a maxiprep kit.

2.1.5. Making electrocompetent cells

Day 1

1. Thoroughly rinse centrifuge and Pyrex bottles with MilliQ water and autoclave the following:
 - a. 500mL of media (5g tryptone, 2.5g yeast extract, 2g NaCl) in a 2-L flask.
 - b. 2× 500mL MilliQ water in rinsed bottles.
 - c. 300mL 10% (v/v) glycerol (270mL water+37.5g glycerol).
 - d. 500mL centrifuge bottle.
 - e. 1.5mL microcentrifuge tubes (~20).
2. Start a 5mL overnight culture from a colony of the desired *E. coli* strain. For bacterial display library construction, we use *E. coli* strain MC1061.

Day 2

1. Refrigerate the bottles of water and glycerol and freeze the centrifuge bottle and labeled microcentrifuge tubes at –20°C.
2. Add 0.2% (w/v) sterile-filtered glucose to 500mL of media and inoculate with 5mL of overnight culture.
3. Grow at 37°C until the OD₆₀₀ reaches 0.6. This occurs after 2–2.5h for the MC1061 strain.
4. Chill the flask in an ice water bath for 30 min. For the remainder of the steps, it is important to keep the cells on ice.
5. Transfer the cells to the cold 500mL centrifuge bottle and centrifuge the cells at 3000×g for 20min at 4°C.

6. Pour off the supernatant being careful to not lose any of the cell pellet. Add 5–10 mL of ice cold sterile MilliQ water and resuspend the pellet by swirling the centrifuge bottle in an ice bath. Then, add the remaining water (total volume of 500 mL) and centrifuge at $3000\times g$ for 20 min at 4°C .
7. Repeat step 6.
8. Pour off the supernatant and add 5–10 mL of cold 10% glycerol and resuspend the pellet as before. Fill the bottle halfway ($\sim 250\text{ mL}$) with 10% glycerol and centrifuge at $3000\times g$ for 10 min at 4°C .
9. Pour off as much liquid as possible without disturbing the pellet and gently resuspend the cell pellet in the remaining glycerol by swirling the centrifuge bottle in an ice bath ($\sim 1.5\text{ mL}$ of total liquid).
10. Aliquot $70\mu\text{L}$ of cell suspension in 15–20 of the cold 1.5 mL centrifuge tubes and freeze immediately at -80°C . For library construction, divide into $250\mu\text{L}$ aliquots with a cell density of approximately $2\times 10^{11}\text{ cells/mL}$.

2.1.6. Measurement of cell competency

1. Thaw an aliquot of electrocompetent cells on ice and add 10 pg of pUC18.
2. Transfer to a cold 1 mm electroporation cuvette and pulse at 1.8 kV , $50\mu\text{F}$, and 100Ω .
3. Rinse the cuvette twice with 0.75 mL of SOC media and grow at 37°C for 1 h.
4. Plate $100\mu\text{L}$ of $1:10^2$ and $1:10^3$ dilutions of the transformation on LB-agar plates supplemented with $100\mu\text{g/mL}$ ampicillin.
5. Grow overnight at 37°C and count the colonies to determine the transformation efficiency of the electrocompetent cells. A count of 100 colonies on platings of the $1:10^2$ dilution corresponds to a transformation efficiency of $1.5\times 10^{10}\text{ CFU}/\mu\text{g pUC18}$.

2.2. Sorting bacterial display libraries to identify peptides that bind a target protein

Below is a general protocol for labeling peptide-displaying bacteria with a biotinylated target protein. The growth and labeling conditions specified are for the eCPX display scaffold. Since eCPX enables presentation of peptides on either scaffold terminus, an affinity tag can be fused to the C-terminus to enable improved resolution of scaffold expression and binding affinity (Kenrick and Daugherty, 2010). For libraries with a diversity greater than 10^7 members, a magnetic selection (MACS) is used to pre-enrich bacteria which display peptides that bind a target protein and to deplete binders to the secondary label streptavidin. One or two rounds of MACS will reduce the

population to a reasonable size for sorting by flow cytometry. Detailed protocols for library screening using MACS and FACS have been described elsewhere ([Kenrick *et al.*, 2007](#)).

1. Grow library population or single clone overnight at 37°C in LB/CM supplemented with 0.2% (w/v) sterile-filtered glucose (glucose is not required for single clones).
2. Subculture at a 1:50 dilution into 5 mL LB/CM and grow for 2 h at 37°C.
3. Induce for 1 h at 37°C with 0.04% (w/v) L-arabinose.
4. If working with a library population, determine the amount of cells needed to oversample the calculated library size by 20- to 50-fold. Be sure to use at least 10 μ L of cells so that a cell pellet is easily visible following centrifugation.
5. Centrifuge the cells at 3000 $\times$ *g* for 5 min and remove the supernatant.
6. Resuspend the pellet in 100 μ L of the biotinylated target protein solution in PBS supplemented with 0.25% (w/v) bovine serum albumin (BSA). In general, a protein concentration of 100–500 nM is a good starting point for the first round of sorting. The concentration of target can be lowered to increase the stringency in later rounds of sorting. Also, prepare unlabeled cells as a negative control.
7. Incubate cells with the target protein at 4°C for 45 min, centrifuge the cells at 3000 $\times$ *g* for 5 min, and remove the supernatant. *Note:* To favor specificity for the target, unlabeled protein mixtures representing diverse specificities (e.g., purified human IgG) can be added before or during this step ([Hall and Daugherty, 2009](#)).
8. Resuspend the pellet in 100 μ L of 15 nM streptavidin-*R*-phycoerythrin (SAPE) solution and incubate at 4°C for 45 min. This step is not required if the protein is directly labeled with a fluorescent dye. Also, a cell sample labeled with SAPE only (skip target protein incubation step) is an important control to test for streptavidin-binding peptides. Neutravidin-phycoerythrin or a fluorescently labeled anti-biotin antibody can be used as an alternative secondary detection agent if streptavidin-binding peptides become enriched during sorting.
9. Centrifuge the cells at 3000 $\times$ *g* for 5 min, remove the supernatant, and resuspend in PBS (500 μ L) for analysis or sorting on the flow cytometer.
10. Load the unlabeled cells (negative control) and adjust the forward scatter (FSC) and side scatter (SSC) detector voltages until the entire cell population is clearly visible on the plot.
11. For the phycoerythrin dye, the flow cytometer should be equipped with a 488-nm laser. Adjust the 576-nm PMT voltage until the mean cell fluorescence of the negative control is at a background level of $\sim$ 100.
12. On the fluorescence plot, set the sort gate so that it encompasses 0.1–1% of the negative control population. Also, adjust the gate to exclude as many streptavidin-binding peptides as possible.

13. Analyze the remaining samples and proceed to sorting (collect the sorted cells in SOC). Oversample the library population by 5- to 10-fold to ensure representation of all the members.

2.3. Troubleshooting the library sorting

The failure to enrich binders to the target can result from a number of potential causes. If library enrichment of binding clones does not occur, try increasing the concentration of target protein and/or re-sort using a population isolated from an earlier round of selection. Also, determine whether the target is properly biotinylated using a degree of labeling kit (optimal degree of labeling is $\sim 2\text{--}4$ mol biotin per mol target protein) or with bacteria displaying a known protein-binding peptide, if available. If possible, analyze the positive control bacteria *after* sorting the library population to prevent contamination with the positive control clone; alternatively, run bleach through the system following the control clone to sanitize the tubing and rinse thoroughly with water prior to sorting.

Avidity and rebinding effects can dramatically increase apparent affinities measured on the cell surface when working with homo-multimeric target proteins. To reduce the avidity effects, the induction time can be shortened (minimum of 10 min) to reduce the number of peptides displayed on the bacterial surface. To diminish rebinding effects, a decoy protein that binds to a tag on the C-terminus of eCPX can increase the steric hindrance and prevent the target protein from unbinding and then reattaching to a neighboring peptide ([Kenrick and Daugherty, 2010](#)). Finally, when measuring dissociation rate constants, as discussed in the next section, an excess concentration of unlabeled competitor can be used so that once a protein dissociates from the cell surface, it is blocked from rebinding. This strategy is especially useful in later rounds of sorting with a multimeric target for isolating peptides that have slow off rates and therefore higher affinity.

One common problem is the enrichment of peptides that bind to the secondary fluorescent probe. This is especially an issue when using streptavidin, which has four biotin-binding sites that can promote tight binding to multiple peptides on the bacterial surface. The number of streptavidin binders can be reduced through depletion sorting for nonbinding clones to the secondary label, or by using streptavidin-coated magnetic beads to remove streptavidin-binding clones. Also, the secondary label can be alternated between streptavidin-phycoerythrin and either neutravidin-phycoerythrin or a fluorescently tagged anti-biotin antibody in each cycle of sorting. If binding to the secondary label remains a problem, eliminate the secondary labeling step and directly conjugate a fluorescent dye (e.g., AlexaFluor 488) to the target protein. When using protein targets conjugated with a fluorophore, the mean cell fluorescence of the binding clones will be significantly lower compared to that obtained by labeling with streptavidin-phycoerythrin.

2.4. Measuring the binding affinity of displayed peptides

Flow cytometry can be used to determine apparent equilibrium dissociation constants (K_d^{APP}) or dissociation rate constants (k_{off}) for peptides displayed on the cell surface (Fig. 4.4). Dissociation constants are to be considered apparent, since they may differ from K_d values measured in solution due to many factors including avidity and rebinding effects. To measure K_d^{APP} , the cells are labeled with a concentration series of target protein (three- to five-fold titration series). The normalized mean cell fluorescence is plotted as a function of the protein concentration (logarithmic scale), and the data can be fit to Eq. (4.2) to determine the K_d^{APP} :

$$\frac{FL - FL_{bg}}{FL_{max} - FL_{bg}} = \frac{[T]}{[T] + K_d^{APP}} \quad (4.2)$$

FL is the mean cell fluorescence of the bacterial cell population, FL_{bg} is the background cell fluorescence of unlabeled bacteria, FL_{max} is the maximum observed fluorescence assuming that every peptide is bound to a target molecule, $[T]$ is the target protein concentration, and K_d^{APP} is the apparent or cell-surface equilibrium dissociation constant (Boder and Wittrup, 1998).

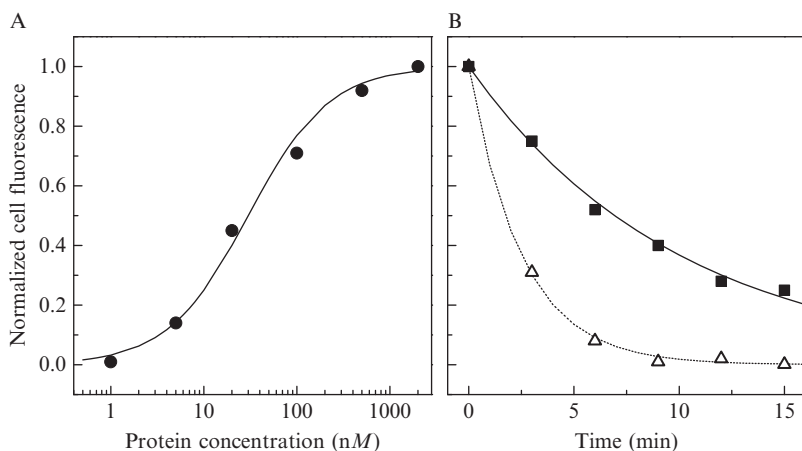


Figure 4.4 Measurement of binding kinetics and affinity. (A) Equilibrium-binding isotherm (hypothetical data) with the normalized cell fluorescence plotted versus the target protein concentration. The data are fit to Eq. (4.2) to calculate the K_d^{APP} for a clonal population of peptide-displaying bacteria. (B) Off-rate analysis (hypothetical data) of two different clones fit to Eq. (4.3) to calculate apparent k_{off} values. A high concentration of unlabeled competitor is added at the initial time point to prevent rebinding of the target protein to the displayed peptide and therefore allows for a more accurate representation of the solution behavior of the peptide.

Another method for ranking individual clones is to determine their relative dissociation rate constants (off rates or k_{off}). Assuming the peptides have similar association rates, their affinity will be strongly influenced by how quickly the target dissociates from the peptide. To quantify the dissociation rate, cells are incubated with saturating concentrations of biotinylated target protein. The cells are then washed once in ice cold buffer and resuspended in an excess of unlabeled competitor to prevent the rebinding of the biotinylated protein to the displayed peptides. At numerous time points, an aliquot is removed and labeled with streptavidin–phycoerythrin for analysis on the flow cytometer. By plotting the normalized cell fluorescence versus time, the dissociation rate constant can be determined using Eq. (4.3):

$$\frac{FL - FL_{\text{bg}}}{FL_0 - FL_{\text{bg}}} = e^{-k_{\text{off}}t} \quad (4.3)$$

FL is the observed mean cell fluorescence, FL_0 is the initial mean cell fluorescence, FL_{bg} is the background cell fluorescence after complete dissociation, t is the time following addition of the unlabeled competitor, and k_{off} is the dissociation rate constant (Boder and Wittrup, 1998).

2.5. Peptide affinity maturation

The affinity of peptides isolated from libraries is typically nonoptimal but can be readily improved using bacterial display in conjunction with FACS. When compared to phage display, the use of cell display and FACS can be advantageous to identify optimal screening conditions and to rapidly characterize the improved clones. For affinity maturation, a second-generation library is typically designed based on peptide sequences identified in the initial screen. The second-generation library is sorted using increased stringency to identify peptide variants with improved affinity for the target. There are several strategies to affinity mature peptides; for example, the focused library can be based on a consensus motif found from the initial library sorting or can be a soft randomization of a single binder. Affinity maturation was recently applied to VEGF binders using bacterial surface display by creating focused libraries based on the naïve 15-mer screen (Kenrick and Daugherty, 2010). Complete affinity maturation protocols have been described elsewhere (Fairbrother *et al.*, 1998; Fleming *et al.*, 2005; Levin and Weiss, 2006; Yu and Smith, 1996).

2.6. Identification of cell-binding peptides

To discover peptide ligands that bind specifically to a given cell type, bacterial display libraries can be rendered intrinsically fluorescent by co-expression of GFP intracellularly (Dane *et al.*, 2006, 2009). Thus, the

binding of bacteria to the target cell type can be readily detected by cytometry to enable simple and efficient screening against many different cell types including stem cells (Little *et al.*, 2011). The library can be counter screened against cell lines that have different receptors to improve the specificity of the ligands for a particular cell type. For example, bacterial display was used to identify peptides that specifically recognized human breast cancer cells over normal cells and could also distinguish breast cancer subtypes (Dane *et al.*, 2006, 2009).

To identify cell-binding peptides (Fig. 4.5), a control cell sample containing only the target mammalian cells is needed to properly set the detector voltages to measure background fluorescence levels. In addition, a sample containing only bacteria enables a clear measurement of the green fluorescence signal and enables location of the bacteria on the FSC–SSC plot. A sample of mammalian cells with GFP-expressing bacteria that express only the display scaffold (no displayed peptide) is crucial for both determining the extent to which the bacterial surface interacts with the mammalian cells as well as validating the sort gate. If a high background signal is observed with this negative control, consider adjusting the wash buffer or number of washes to reduce nonspecific binding of the bacteria to the mammalian cells. With an efficient cell-binding peptide, it is possible for 10 or more bacteria to bind to an individual mammalian cell. In addition to cytometry, these samples can be easily imaged using fluorescence microscopy to visualize the interaction of the GFP bacteria with the target cells.

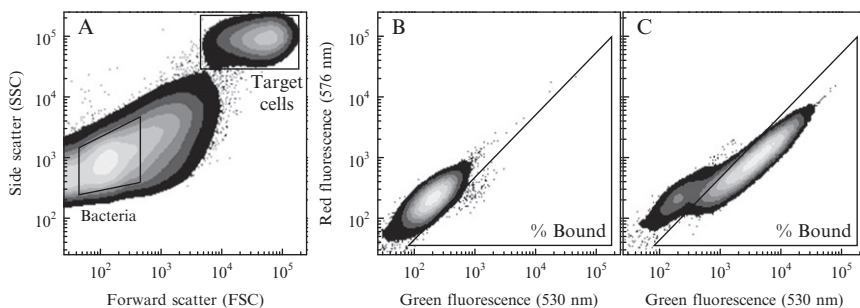


Figure 4.5 Screening for cell-binding peptides. (A) Scatter plot (side vs. forward scatter with both on a logarithmic scale) with the detector voltages adjusted so that the bacteria and target cells are two complete and distinct populations. (B) Bacteria expressing GFP, but without a displayed peptide, show negligible binding to the target cells. This plot only displays events within the target cell gate; the 576-nm fluorescence is due to GFP cross talk with the 576-nm channel (only one fluorophore is present in this system). (C) Bacteria expressing a receptor-specific peptide bind efficiently to the tumor cells as shown by the large increase in green fluorescence. The events in the % bound gate represent target cells that have one or more bacteria bound to their surface, and the majority of tumor cells are labeled with green bacteria due to the peptide ligand.

1. Follow the general growth and induction procedures (steps 1–4) in the Target Protein Sorting protocol (Section 2.2).
2. While the bacteria are growing, harvest the mammalian cells from dishes by scraping, or by using another trypsin-independent dissociation method, to preserve the integrity of cell-surface receptors. Be sure to rinse off any media containing antibiotic using PBS.
3. Count the target cells using a hemocytometer. The cell suspension should have >90% viability as determined by the trypan blue cell-viability test.
4. Take an aliquot of the cell suspension adequate to obtain between 2 and 5×10^5 cells per sample, and spin down the cells in a clinical centrifuge at $150 \times g$ for 7 min.
5. Resuspend the cell pellet in PBS (50 μ L for each sample).
6. After induction, dilute the bacteria 1:2 and measure the OD₆₀₀; assume a value of 1 corresponds to 1×10^9 bacterial cells/mL. Determine the volume of bacteria needed for a ratio of 50:1 bacteria to mammalian cells ($1\text{--}2.5 \times 10^7$ bacteria/sample) and aliquot into separate tubes for each sample.
7. Centrifuge the bacteria at $3000 \times g$ for 5 min and resuspend the pellet with 50 μ L of the mammalian cell suspension.
8. Incubate the mammalian cells and bacteria together for 1 h at 4°C on an inversion shaker.
9. Wash the sample three times with PBS:
 - a. Add 1 mL of PBS and pipet up and down to mix.
 - b. Centrifuge the cells at $1500 \times g$ for 30 s using a swinging bucket rotor. The mammalian cells should form a pellet, while most unbound bacteria remain in the supernatant.
 - c. Remove the supernatant.
 - d. Repeat steps (a) through (c) twice.
10. Resuspend the cell pellet in PBS (250 μ L) and filter the samples using 35 μ m cell-strainer caps (BD Falcon).
11. Load the target cells only and bacteria only control samples, and adjust the FSC and SSC voltages so that both the bacteria and mammalian cell populations are clearly visible on the scatter plot.
12. Plot 576 nm (red) versus 530 nm (green) fluorescence of the target cell population with the graph only displaying events from the target cell gate.
13. Adjust the 530 and 576 nm detector voltages so that the mean fluorescence of the target cell population is set at the background level (~ 100). Also, draw the sort gate (% bound in Fig. 4.5B) on the 576 versus 530 nm fluorescence plot so that only $\sim 0.1\%$ of the target cells are within the gate. The GFP-expressing bacteria should produce green fluorescent signals that are $\sim 5\text{--}10$ -fold above the background autofluorescence of the target cells.

14. Analyze the sample containing target cells with nonbinding fluorescent bacteria (only the display scaffold), and verify that only a small percentage of the cells are within the sort gate. This negative control is important for determining the level of nonspecific binding and for optimizing the wash conditions.
15. Analyze the remaining samples and sort the events in the % bound gate.

2.7. Protease substrate identification using cellular libraries of peptide substrates (CLiPS)

A two-color bacterial display system termed cellular libraries of peptide substrates (CLiPS) was developed to quickly identify novel protease substrates (Boulware *et al.*, 2010, Jabaiah and Daugherty, 2011). For CLiPS, the plasmid pBAD33-eCPX-CLiPS codes a streptavidin-binding peptide (WVCHPMWEVMCLR; Rice *et al.*, 2006) upstream of the substrate library fused to the N-terminus of eCPX (Fig. 4.3B). On the C-terminus of eCPX, a SH3 domain-binding peptide (PAPSIDRSTKPPL; Harkiolaki *et al.*, 2003) was added to verify expression of the full-length construct and enable two color screening for cleaved substrates. The bacteria can be incubated with both SAPE (red probe) and *Alaj*GFP (green probe) fused to the SH3 domain of monocytic adaptor protein (Mona) to separately label both termini of the eCPX construct. For the CLiPS procedure (Fig. 4.6), the selection rounds consist of sorting for bacteria that are efficiently labeled with both the red and green probes at the N- and C-terminus, followed by a round of sorting post-protease treatment where the red fluorescence decreases while the green fluorescence remains high which indicates substrate hydrolysis. Sorting bacteria that are positive for both probes is important to remove substrate peptides that bind to the *Alaj*GFP-Mona fusion protein (green-positive but red-negative). It also removes bacteria that do not properly display the substrate peptide (both green- and red-negative) due to stop codons or frame shift mutations from the library construction.

If a canonical substrate is known, it can make a useful positive control for cleavage and also serve as a comparison for determining the improved cleavage kinetics of the isolated substrates. Also, this substrate can be used to determine the best incubation time and protease concentration for cleavage prior to sorting the library. It is important to pick the best reaction buffer for each protease to ensure maximal activity (usually a Tris/HEPES-based buffer, with cofactors possibly being required).

1. Follow the general growth and induction procedures (steps 1–4) in the Target Protein Sorting protocol (Section 2.2).
2. Incubate the bacteria with protease at room temperature or 37°C for 1–3h (skip this step if sorting for labeling only). Multiple protease

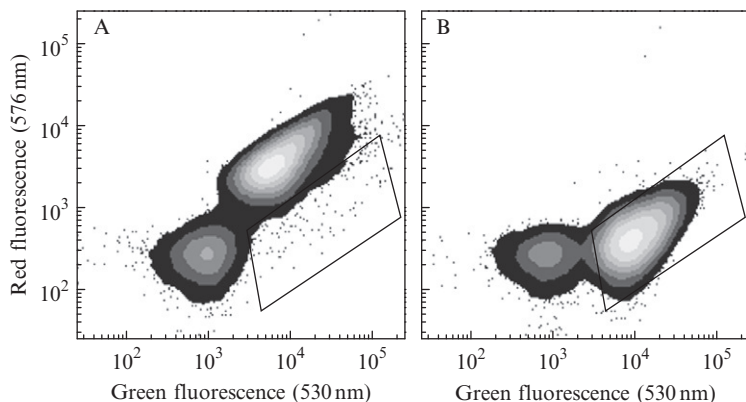


Figure 4.6 Screening for protease substrates. (A) Bacteria that are labeled with a red fluorescent probe (SAPE) which binds to the N-terminus of eCPX and a green fluorescent probe (*Alaj*GFP-Mona) which interacts with the C-terminus. With no protease present, the red- and green-positive cells are sorted to ensure expression of full-length eCPX constructs. (B) The bacteria are incubated with the protease, and the hydrolysis of the substrate causes a large decrease in the red fluorescence (streptavidin-binding peptide is no longer attached to the cell surface) while the green signal remains positive.

concentrations and incubation times can be tested to determine the best conditions for cleavage and sorting.

3. Dilute the sample 100-fold in PBS to stop the reaction, centrifuge the cells at $3000\times g$ for 5 min, and remove the supernatant.
4. Label the bacteria with SAPE (50 nM) and *Alaj*GFP-Mona fusion (250 nM) in PBS ($\sim 100\mu\text{L}$).
5. Incubate at 4°C for 1 h.
6. Centrifuge the cells at $3000\times g$ for 5 min and resuspend the pellet in PBS ($500\mu\text{L}$) for analysis or sorting on the flow cytometer.
7. Load an unlabeled bacteria control to set the 530 and 576 nm detector PMT voltages, so the mean red and green fluorescence is at the background level (~ 100).
8. Analyze the bacteria labeled with both the red and green probes (no protease) and draw the sort gate based on this control with the gate extending to lower red fluorescence (Fig. 4.6).
9. Analyze the remaining samples and sort the cell population with reduced red fluorescence.

3. CONCLUSIONS

The protocols described above enable the construction and screening of large bacterial display libraries. Given the efficiency of peptide library construction and screening in *E. coli*, distinct libraries can be readily tailored to accomplish diverse experimental objectives; these protocols illustrate

methods of peptide ligand engineering for proteins and cell membrane receptors along with protease substrate discovery. In addition, the ability to screen large peptide libraries quantitatively for specific functional properties using FACS provides unique opportunities to adapt the presented techniques for a variety of peptide discovery and design goals. Overall, the flexible nature of bacterial display peptide libraries allows these construction and screening protocols to evolve in order to address a broad range of emerging applications.

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